Lesson plan · for the teacher

Fair or unfair models

Data science · Level 4 · Lesson 3

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Find bias with error rates by group.
- Explain where bias starts.
- Plan checks before adopting a model.

Activity: Audit a model

1. Warm-up (5 min): Read the reading-app story and ask: would you have noticed the problem from the overall score?
2. Teach (8 min): Split a results table by group on the board and work out each rate.
3. Audit (15 min): Groups get results for a made-up tool, work out rates for two groups and list possible causes.
4. Plan (7 min): Each group writes three questions to ask before the school adopts the tool.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Which group carried most errors?
- What in the data could explain it?
- Who should check the results?

Exit ticket

- Group A: 3 wrong in 20. Group B: 9 wrong in 20. Write both error rates.

Try it in class tomorrow
Give groups a made-up table of a model's results for two classes. They work out the error rate for each and decide whether the model is fair.

Worksheet · for pupils

Fair or unfair models

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Name _____ Class _____ Date _____

1 **Where does most bias in a model start?**

◯ In the data it learned from ◯ In the screen colour ◯ In the pupils

2 **A model with high overall accuracy cannot be biased.**

◯ True ◯ False

3 **Finish rate(xs) so it returns the error rate as a whole percentage, then print the rate for each group.**

```
town = [0, 0, 1, 0, 0, 0, 0, 0, 0, 0]
village = [1, 0, 1, 0, 0, 1, 0, 0, 1, 0]

def rate(xs):
    return 0

print(rate(town), rate(village))
```

Write your program:

4 **Which steps make a reading app fairer? Choose all that apply.**

☐ Train it on more varied voices ☐ Have a teacher check low scores

☐ Ask the supplier for error rates by group ☐ Hide the results from teachers

5 **Town pupils: 6 wrong scores per 100. Village pupils: 22 per 100. What does this show?**

◯ The model is unfair to village pupils ◯ Village pupils read worse

◯ The model is fair on average, so all is well

✎ **Which tool or test at your school might work less well for some pupils? How would you find out?**